

Avnet, Miami SoM from Topic Embedded Products and MitySOM from Critical Links.

An interesting example of an FPGA based SoM application is sensor fusion, where inputs from different types of sensors with different interfaces and different communication protocols are processed in real-time. The processor manages all incoming data sequentially and the programmable IC processes all these inputs in parallel in real-time.

Packaging. The density of an IC package is growing. Bigger designs are made possible with smaller chips. With increased density, the number of chips required reduces. This reduces interconnectivity, thereby decreasing the complexity of the entire design, increasing flexibility and offering better clock timings.

Seamless migration. One feature programmable IC firms offer is easy migration between different models of the same brand. This gives the user the possibility to upgrade from or replace an older, low-end version to the latest, mid-range or high-end version of FPGA.

Enabling tools

Major enhancements have been made in software tools for programmable devices. The software is now available with numerous built-in features that save a designer's time in repeating the design process. Such advanced tools take care of so many factors that make the software much easier to use and increase the designers' productivity by better reporting and increased performance, including common connectivity blocks to accelerate development cycle. Software flow for programmable devices is also becoming more flexible by allowing designers to adopt their own flow or architectures, like MATLAB/Simulink, ARM DS5, OpenCL and high-level synthesis (HLS).

The HLS of Xilinx Vivado design suite allows the user to write programs for ASICs and FPGAs in C and C++ languages, and easily convert

Comparison between CPLDs and FPGAs

Complex programmable logic devices (CPLDs)	Field programmable gate arrays (FPGAs)
Non-volatile	Volatile
Used in devices that require small gate count	Used in devices that require high gate count
Less complex architecture	More complex architecture
Less cost per package	High cost per package
Logic cells are based on EEPROM flash	Designed using RAM based digital logic chip
Coarse grain type: a few number of large logic blocks	Fine grain type: a large number of tiny logic blocks
Delays are much predictable	Delays are quite unpredictable
For mass markets and small, simple control based designs that require quick response time, fast decoding, ultra-low idle power consumption and design security. For example, battery-operated equipment	Covers a wide range from low-cost, low-end models for simple applications to optimum cost, mid-range models for slightly complex applications to high-cost, high-end models for highly complex applications

these to hardware description languages (HDLs). So, even software engineers can now design with FPGAs. This is a big productivity boost for customers, as many people know C and C++ but not HDL, VHDL and Verilog. Software tools, like software-defined specification environment for networking (SDNet) and SDAccel by Xilinx, are software development environments for FPGAs and/or SoCs. These tools help in better hardware and software partitioning and improve design integrity.

The DesignInsight technology by Tabula, which consists of architecture, chip design and Stylus compiler software, allows real-time observation, debugging and validation of individual logic cells of PLD in real-time (up to 2GHz).

Application and industry specific

FPGAs are programmable for all industries. But there are several stringent certification procedures and requirements to approve a device for certain applications, like military, automotive and medical. Due to this reason major FPGA manufacturers are now introducing FPGAs specific to certain fields of applications. These are becoming more ASIC-like. For example, Altera Enpirion PowerSoCs and Cyclone V SoC FPGAs are automotive grade, Altera Arria 10 FPGAs and SoCs are fully-compliant with military standards, and Kintex

Ultrascale is suited for data centres, video and medical imaging, broadcast systems and radars.

Data centre acceleration. Apart from working closely with CPUs and improving response time, throughput and efficiency, FPGAs these days also make programming and debugging easy by letting designers work in languages and architectures they are familiar with, such as MATLAB/Simulink, OpenCL, ARM DS-5 and HLS. This is a major advantage to data centre programmers.

Network enrichment. The IoT environment is varied and uncertain, mentions Steve Gabriel, corporate communications manager at Altera Corp. The fast, computationally efficient programmable chips are suitable for the IoT segment in terms of power efficiency, time sensitivity and security. These are effective for IoT hubs that have to support a large number of ultra-low-power network protocols, constantly evolving safety and security tasks, and a wide variety of application-dependent, real-time computing loads.

Real-time 3D medical imaging. Medical image processing and analysis are computationally-intensive tasks and require larger memories. The use of programmable chips, like high-powered CPUs and FPGAs, allow real-time, high-speed and accurate processing and analysis of photonic and electronic signals, with high image resolution. It is also